

S.Y. (B.Tech) Semester III
Subject: Statistics for Data Science
Experiment 9

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Date:	Experiment Title: Analysis of Variance using F-distribution																																																
Aim	To implement one-way and two-way ANOVA																																																
Software	Google Colab																																																
Theory	1. Can we conclude that treatments are effective on plant growth from the given data? Also check the significance using Tukey’s Test. <pre>In [2]: import numpy as np import pandas as pd from statsmodels.formula.api import ols from statsmodels.stats.multicomp import pairwise_tukeyhsd import statsmodels.api as sm</pre><pre>In [3]: from google.colab import drive drive.mount('/content/drive')</pre> Mounted at /content/drive <pre>In [5]: path = "/content/drive/MyDrive/SDS dataset/Plant_Growth_New.csv" df = pd.read_csv(path) df</pre><pre>In [5]: path = "/content/drive/MyDrive/SDS dataset/Plant_Growth_New.csv" df = pd.read_csv(path) df</pre> Out[5]: <table><tr><th></th><th>Treatment</th><th>Height</th></tr><tr><td>0</td><td>low sun exposure and water daily</td><td>6</td></tr><tr><td>1</td><td>low sun exposure and water daily</td><td>6</td></tr><tr><td>2</td><td>low sun exposure and water daily</td><td>6</td></tr><tr><td>3</td><td>low sun exposure and water daily</td><td>5</td></tr><tr><td>4</td><td>low sun exposure and water daily</td><td>6</td></tr><tr><td>5</td><td>medium sun exposure and water daily</td><td>5</td></tr><tr><td>6</td><td>medium sun exposure and water daily</td><td>5</td></tr><tr><td>7</td><td>medium sun exposure and water daily</td><td>6</td></tr><tr><td>8</td><td>medium sun exposure and water daily</td><td>4</td></tr><tr><td>9</td><td>medium sun exposure and water daily</td><td>5</td></tr><tr><td>10</td><td>high sun exposure and water daily</td><td>6</td></tr><tr><td>11</td><td>high sun exposure and water daily</td><td>6</td></tr><tr><td>12</td><td>high sun exposure and water daily</td><td>7</td></tr><tr><td>13</td><td>high sun exposure and water daily</td><td>8</td></tr><tr><td>14</td><td>high sun exposure and water daily</td><td>7</td></tr></table>		Treatment	Height	0	low sun exposure and water daily	6	1	low sun exposure and water daily	6	2	low sun exposure and water daily	6	3	low sun exposure and water daily	5	4	low sun exposure and water daily	6	5	medium sun exposure and water daily	5	6	medium sun exposure and water daily	5	7	medium sun exposure and water daily	6	8	medium sun exposure and water daily	4	9	medium sun exposure and water daily	5	10	high sun exposure and water daily	6	11	high sun exposure and water daily	6	12	high sun exposure and water daily	7	13	high sun exposure and water daily	8	14	high sun exposure and water daily	7
	Treatment	Height																																															
0	low sun exposure and water daily	6																																															
1	low sun exposure and water daily	6																																															
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8	medium sun exposure and water daily	4																																															
9	medium sun exposure and water daily	5																																															
10	high sun exposure and water daily	6																																															
11	high sun exposure and water daily	6																																															
12	high sun exposure and water daily	7																																															
13	high sun exposure and water daily	8																																															
14	high sun exposure and water daily	7																																															

15	low sun exposure and water weekly	3
16	low sun exposure and water weekly	4
17	low sun exposure and water weekly	4
18	low sun exposure and water weekly	4
19	low sun exposure and water weekly	5
20	medium sun exposure and water weekly	4
21	medium sun exposure and water weekly	4
22	medium sun exposure and water weekly	4
23	medium sun exposure and water weekly	4
24	medium sun exposure and water weekly	4
25	high sun exposure and water weekly	5
26	high sun exposure and water weekly	6
27	high sun exposure and water weekly	6
28	high sun exposure and water weekly	7
29	high sun exposure and water weekly	8

```
print("Null Hypothesis is that tthere is no significant difference between the treatment on plants with their growth")
print("Alternate Hypothesis is that tthere is significant difference between the treatment on plants with their growth")
alpha=0.05
mod=ols("Height~Treatment",data=df).fit()
aov=sm.stats.anova_lm(mod)
print("One way Anova Table is given below:\n",aov)
```

Null Hypothesis is that tthere is no significant difference between the treatment on plants with their growth
 Alternate Hypothesis is that tthere is significant difference between the treatment on plants with their growth
 One way Anova Table is given below:

	df	sum_sq	mean_sq	F	PR(>F)
Treatment	5.0	35.866667	7.173333	13.45	0.000003
Residual	24.0	12.800000	0.533333	NaN	NaN

Null rejected

```
f_statistics=aov['F']['Treatment']
p_value=aov['PR(>F)']['Treatment']
print('f_statistics=',f_statistics,'& p_value=',p_value)
if p_value>alpha:
    print("Failed to reject null hypothesis for level of significance= "+str(alpha))
else:
    print("Null hypothesis is rejected for level of significance= "+str(alpha))
```

f_statistics= 13.449999999999998 & p_value= 2.6191797394662647e-06
 Null hypothesis is rejected for level of significance= 0.05

```
tukey=pairwise_tukeyhsd(endog=df['Height'],groups=df['Treatment'],alpha=0.05)
print("Tukey Post-Hoc analysis for the given dataset is given below:\n",tukey)
```

Tukey Post-Hoc analysis for the given dataset is given below:

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
high sun exposure and water daily	high sun exposure and water weekly	-0.4	0.9	-1.8282	1.0282	False
high sun exposure and water daily	low sun exposure and water daily	-1.0	0.29	-2.4282	0.4282	False
high sun exposure and water daily	low sun exposure and water weekly	-2.8	0.001	-4.2282	-1.3718	True
high sun exposure and water daily	medium sun exposure and water daily	-1.8	0.0079	-3.2282	-0.3718	True
high sun exposure and water daily	medium sun exposure and water weekly	-2.8	0.001	-4.2282	-1.3718	True
high sun exposure and water weekly	low sun exposure and water daily	-0.6	0.7576	-2.0282	0.8282	False
high sun exposure and water weekly	low sun exposure and water weekly	-2.4	0.001	-3.8282	-0.9718	True
high sun exposure and water weekly	medium sun exposure and water daily	-1.4	0.057	-2.8282	0.0282	False
high sun exposure and water weekly	medium sun exposure and water weekly	-2.4	0.001	-3.8282	-0.9718	True
low sun exposure and water daily	low sun exposure and water weekly	-1.8	0.0079	-3.2282	-0.3718	True
low sun exposure and water daily	medium sun exposure and water daily	-0.8	0.5222	-2.2282	0.6282	False
low sun exposure and water daily	medium sun exposure and water weekly	-1.8	0.0079	-3.2282	-0.3718	True
low sun exposure and water weekly	medium sun exposure and water daily	1.0	0.29	-0.4282	2.4282	False
low sun exposure and water weekly	medium sun exposure and water weekly	0.0	0.9	-1.4282	1.4282	False
medium sun exposure and water daily	medium sun exposure and water weekly	-1.0	0.29	-2.4282	0.4282	False

2. A botanist wants to know whether or not plant growth is influenced by sunlight exposure and watering frequency. She plants 30 seeds and lets them grow for two months under different conditions for sunlight exposure and watering frequency. After two months, she records the height of each plant, in inches. Perform a two-way ANOVA to determine if watering frequency and sunlight exposure have a significant effect on plant growth, and to determine if there is any interaction effect between watering frequency and sunlight exposure.

- 5 plants were given low sun exposure and water daily
- 5 plants were given medium sun exposure and water daily
- 5 plants were given high sun exposure and water daily
- 5 plants were given low sun exposure and water weekly
- 5 plants were given medium sun exposure and water weekly
- 5 plants were given high sun exposure and water weekly

Height of plant: 6, 6, 6, 5, 6, 5, 5, 6, 4, 5, 6, 6, 7, 8, 7, 3, 4, 4, 4, 5, 4, 4, 4, 4, 4, 5, 6, 6, 7, 8

```
path = "/content/drive/MyDrive/SDS dataset/Plant_Growth_New_1"
df = pd.read_csv(path)
df
```

	Sun_Exposure	Water	Height
0	low	daily	6
1	low	daily	6
2	low	daily	6
3	low	daily	5
4	low	daily	6
5	medium	daily	5
6	medium	daily	5
7	medium	daily	6
8	medium	daily	4
9	medium	daily	5
10	high	daily	6
11	high	daily	6

12	high	daily	7
13	high	daily	8
14	high	daily	7
15	low	weekly	3
16	low	weekly	4
17	low	weekly	4
18	low	weekly	4
19	low	weekly	5
20	medium	weekly	4
21	medium	weekly	4
22	medium	weekly	4
23	medium	weekly	4
24	medium	weekly	4
25	high	weekly	5
26	high	weekly	6
27	high	weekly	6
28	high	weekly	7
29	high	weekly	8

```
print("Null Hypothesis H_01 is that there is no significant effect of Sunlight Exposure on plant growth.")
print("Alternate Hypothesis H_a1 is that there is significant effect of Sunlight Exposure on plant growth.")
print("Null Hypothesis H_02 is that there is no significant effect of Watering Frequency on plant growth.")
print("Alternate Hypothesis H_a2 is that there is significant effect of Watering Frequency on plant growth.")
print("Null Hypothesis H_03 is that there is no significant interaction effect of Watering Frequency & Sunlight exposure on plant growth.")
print("Alternate Hypothesis H_a3 is that there is significant interaction effect of Watering Frequency & Sunlight exposure on plant growth.")
```

Null Hypothesis H_01 is that there is no significant effect of Sunlight Exposure on plant growth.
 Alternate Hypothesis H_a1 is that there is significant effect of Sunlight Exposure on plant growth.
 Null Hypothesis H_02 is that there is no significant effect of Watering Frequency on plant growth.
 Alternate Hypothesis H_a2 is that there is significant effect of Watering Frequency on plant growth.
 Null Hypothesis H_03 is that there is no significant interaction effect of Watering Frequency & Sunlight exposure on plant growth.
 Alternate Hypothesis H_a3 is that there is significant interaction effect of Watering Frequency & Sunlight exposure on plant growth.

```
mod=ols('Height~Sun_Exposure*Water',data=df).fit()
aov=sm.stats.anova_lm(mod)
print("Two way Anova Table is given below:\n",aov)
```

Two way Anova Table is given below:

	df	sum_sq	mean_sq	F	PR(>F)
Sun_Exposure	2.0	24.866667	12.433333	23.3125	0.000002
Water	1.0	8.533333	8.533333	16.0000	0.000527
Sun_Exposure:Water	2.0	2.466667	1.233333	2.3125	0.120667
Residual	24.0	12.800000	0.533333	NaN	NaN

	Null rejected <pre>f_statistics_1=aov['F']['Sun_Exposure'] p_value_1=aov['PR(>F)']['Sun_Exposure'] print('f_statistics_1=',f_statistics_1,'& p_value_1=',p_value_1) if p_value_1>alpha: print("Failed to reject null hypothesis H_01 for level of significance= "+str(alpha)) else: print("Null hypothesis H_01 is rejected for level of significance= "+str(alpha))</pre> f_statistics_1= 23.312500000000004 & p_value_1= 2.371555925858255e-06 Null hypothesis H_01 is rejected for level of significance= 0.05 <pre>f_statistics_2=aov['F']['Water'] p_value_2=aov['PR(>F)']['Water'] print('f_statistics_2=',f_statistics_2,'& p_value_2=',p_value_2) if p_value_2>alpha: print("Failed to reject null hypothesis H_02 for level of significance= "+str(alpha)) else: print("Null hypothesis H_02 is rejected for level of significance= "+str(alpha))</pre> f_statistics_2= 16.0 & p_value_2= 0.0005269080727817035 Null hypothesis H_02 is rejected for level of significance= 0.05 <pre>f_statistics_3=aov['F']['Sun_Exposure:Water'] p_value_3=aov['PR(>F)']['Sun_Exposure:Water'] print('f_statistics_3=',f_statistics_3,'& p_value_3=',p_value_3) if p_value_3>alpha: print("Failed to reject null hypothesis H_03 for level of significance= "+str(alpha)) else: print("Null hypothesis H_03 is rejected for level of significance= "+str(alpha))</pre>
Conclusion	Hence we have studied and implemented analysis of variance using F-distribution.

Signature of Faculty